

6.9) a) START WITH $i = j$

$$\Rightarrow q_i \cdot q_i = \frac{u_i \cdot u_i}{|u_i| |u_i|} = \frac{|u_i|^2}{|u_i|^2} = 1$$

CONSIDER $i \neq j$ AND PROVE $q_i \cdot q_j = 0$

$$q_i \cdot q_j = q_i \cdot \left(a_j - \sum_{k=0}^{j-1} (q_k \cdot a_j) q_k \right) \frac{1}{|u_j|}$$

$$= C \left[q_i \cdot a_j - \sum_{k=0}^{j-1} (q_k \cdot a_j) q_i \cdot q_k \right]$$

$$C = 1/|u_j|$$

CHECK $i=0, j=1$

$$q_0 \cdot q_1 = C [q_0 \cdot a_1 - (q_0 \cdot a_1)(q_0 \cdot q_0)] = 0$$

CHECK $i=0, j=2$

$$q_0 \cdot q_2 = C [q_0 \cdot a_2 - (q_0 \cdot a_2)(q_0 \cdot q_0) - (q_1 \cdot a_2)(q_1 \cdot q_0)] = 0$$

ASSUME q_j ARE ORTHONORMAL UP TO $j=n$

$$q_i \cdot q_{n+1} = C \left[q_i \cdot a_{n+1} - \sum_{k=0}^n (q_k \cdot a_{n+1}) q_i \cdot q_k \right]$$
$$= C [q_i \cdot a_{n+1} - q_i \cdot a_{n+1}] = 0$$

$\Rightarrow q_j$ ARE ORTHONORMAL UP TO $j=n+1$

$\Rightarrow g_i$ ARE ORTHONORMAL FOR ALL n

$$\Rightarrow g_i \cdot g_j = \begin{cases} 1 & \text{IF } i=j \\ 0 & \text{IF } i \neq j \end{cases}$$

4) CREATE MATRIX A

DEF QR(A):

INITIALIZE Q (SHAPE OF A)

INITIALIZE R (SHAPE OF A)

FOR $i=0 \dots \text{NUM COLUMNS IN } A$:

$$Q[:, i] = A[:, i]$$

FOR $j=0 \dots i$:

$$\text{PROJECTION} = Q[:, j] \cdot A[:, i]$$

$$Q[:, i] -= \text{PROJECTION} \times Q[:, j]$$

$$R[j, i] = \text{PROJECTION}$$

$$\text{MAGNITUDE} = \text{MAG}(Q[:, i])$$

$$R[i, i] = \text{MAGNITUDE}$$

$$Q[:, i] /= \text{MAGNITUDE}$$

$$6.10) a) \hat{H} = -\frac{\hbar^2}{2M} \frac{d^2}{dx^2} + V(x)$$

$$\hat{H} \psi(x) = E \psi(x)$$

$$\Rightarrow \hat{H} \sum_{n=1}^{\infty} \psi_n \sin\left(\frac{n\pi x}{L}\right) = E \sum_{n=1}^{\infty} \psi_n \sin\left(\frac{n\pi x}{L}\right)$$

$$\begin{aligned} \Rightarrow \sin\left(\frac{m\pi x}{L}\right) \hat{H} \sum_{n=1}^{\infty} \psi_n \sin\left(\frac{n\pi x}{L}\right) \\ = \sin\left(\frac{m\pi x}{L}\right) E \sum_{n=1}^{\infty} \psi_n \sin\left(\frac{n\pi x}{L}\right) \end{aligned}$$

INTEGRATE BOTH SIDES

$$\text{LHS} = \sum_{n=1}^{\infty} \int_0^L \sin\left(\frac{m\pi x}{L}\right) \hat{H} \psi_n \sin\left(\frac{n\pi x}{L}\right) dx$$

$$\text{RHS} = \sum_{n=1}^{\infty} \int_0^L \psi_n E \sin\left(\frac{m\pi x}{L}\right) \sin\left(\frac{n\pi x}{L}\right) dx$$

ONLY THE $n=m$ CONTRIBUTES TO THE SUM.

$$\Rightarrow \text{RHS} = \frac{L}{2} E \psi_m$$

$$\Rightarrow \sum_{n=1}^{\infty} \psi_n \int_0^L \sin\left(\frac{m\pi x}{L}\right) \hat{H} \sin\left(\frac{n\pi x}{L}\right) dx = \frac{1}{2} L E \psi_m$$

$$\Rightarrow \sum_{n=1}^{\infty} \psi_n \frac{L}{2} H_{mn} = \frac{L}{2} E \psi_m$$

$$\Rightarrow \sum_{n=1}^{\infty} \psi_n H_{mn} = E \psi_m \quad \text{THIS IS JUST A MATRIX MULT.}$$

$$\Rightarrow \vec{H} \vec{\Psi} = E \vec{\Psi}$$

$$1) H_{mm} = \frac{2}{L} \int_0^L \sin\left(\frac{m\pi x}{L}\right) \left[-\frac{\hbar^2}{2M} \frac{d^2}{dx^2} + \frac{ax}{L} \right] \sin\left(\frac{n\pi x}{L}\right) dx$$

BREAK INTO 2:

$$I_1 = \frac{-\hbar^2}{LM} \int_0^L \sin\left(\frac{m\pi x}{L}\right) \frac{d^2}{dx^2} \sin\left(\frac{n\pi x}{L}\right) dx$$

$$= \frac{\hbar^2}{LM} \left(\frac{n\pi}{L}\right)^2 \int_0^L \sin\left(\frac{m\pi x}{L}\right) \sin\left(\frac{n\pi x}{L}\right) dx$$

$$= \frac{\hbar^2}{LM} \left(\frac{n\pi}{L}\right)^2 \frac{L}{2} \quad \text{IF } m=n$$

$$= \frac{n^2 \pi^2 \hbar^2}{2L^3 M} \quad \text{IF } m=n$$

$$I_2 = \frac{2}{L} \int_0^L \sin\left(\frac{m\pi x}{L}\right) \frac{ax}{L} \sin\left(\frac{n\pi x}{L}\right) dx$$

$$= \begin{cases} 0 & m \neq n \quad m, n \text{ ARE BOTH EVEN OR ODD} \\ -\frac{8a}{\pi^3} \frac{mn}{(m^2 - n^2)^2} & m \neq n \quad \text{ONE EVEN, ONE ODD} \\ \frac{a}{2} & m = n \end{cases}$$

$$H_{mm} = \begin{cases} 0 \\ -\frac{8a}{\pi^3} \frac{mn}{(m^2 - n^2)^2} \\ I_1 + a/2 \end{cases}$$